less than that of the parent nucleus. The difference in mass ($\Delta m = m_X - m_Y - m_\alpha$) is released as energy called **disintegration energy** Q and is given by

$$Q = \left(m_X - m_Y - m_\alpha\right)c^2 \qquad (9.27)$$

Note that for spontaneous decay (natural radioactivity) Q >0. In alpha decay process, the disintegration energy is certainly positive (Q > 0). In fact, the disintegration energy Q is also the net kinetic energy gained in the decay process or if the parent nucleus is at rest, Q is the total kinetic energy of daughter nucleus and the ${}^{4}_{2}He$ nucleus. Suppose Q < 0, then the decay process cannot occur spontaneously and energy must be supplied to induce the decay.

Note

In alpha decay, why does the unstable nucleus emit ${}^{4}_{2}He$ nucleus? Why it does not emit four separate nucleons? After all ${}^{4}_{2}He$ consists of two protons and two neutrons. For example, if ${}^{238}_{92}U$ nucleus decays into ${}^{234}_{90}Th$ by emitting four separate nucleons (two protons and two neutrons), then the disintegration energy Q for this process turns out to be negative. It implies that the total mass of products is greater than that of parent(${}^{238}_{92}U$) nucleus. This kind of process cannot occur in nature because it would violate conservation of energy. In any decay process, the conservation of energy, laws of linear momentum and laws of angular momentum must be obeyed.

EXAMPLE 9.11

(a) Calculate the disintegration energy when stationary ${}^{232}_{92}U$ nucleus decays to thorium ${}^{228}_{90}Th$ with the emission of α particle. The atomic masses are of ${}^{232}_{92}U = 232.037156\,u$, ${}^{228}_{90}Th = 228.028741\,u$ and ${}^{4}_{2}He = 4.002603\,u$

(b) Calculate kinetic energies of ${}^{228}_{90}Th$ and α-particle and their ratio.

Solution

The difference in masses

$\Delta m = \left(m_U - m_{Th} - m_\alpha\right)$

= (232.037156–228.028741 – 4.002603)u

The mass lost in this decay = 0.005812 u

Since 1u = 931MeV, the energy Q released is

$Q = (0.005812u) \times (931MeV / u)$

$= 5.41\,MeV$

This disintegration energy Q appears as the kinetic energy of α particle and the daughter nucleus.

In any decay, the total linear momentum must be conserved.

Total linear momentum of the parent nucleus = total linear momentum of the daughter nucleus and α particle

Since before decay, the uranium nucleus is at rest, its momentum is zero.

By applying conservation of momentum, we get

$0 = m_{Th}\vec{v}_{Th} + m_\alpha\vec{v}_\alpha$

$m_\alpha\vec{v}_\alpha = -m_{Th}\vec{v}_{Th}$

It implies that the alpha particle and daughter nucleus move in opposite directions.

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